MILITARY SPECIFICATION

WAVEGUIDES, RIGID, RECTANGULAR

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for seamless or fabricated rigid waveguides with rectangular inside configurations (see 6.1).

1.2 Military part number. The military part number shall be as follows:

M	85/1	-	001	-	096
Military designator	Specification sheet number		Arbitary dash number (see 3.1)		Length in inches (see 3.3.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
QQ-S-781	-	Strapping, Steel, Flat and Seals.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape: Packaging Waterproof.
PPP-T-66	-	Tape: Pressure-Sensitive Adhesive, Vinyl Plastic Film.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive, Paper, (For Carton Sealing).
PPP-T-495	-	Tube, Mailing, and Filling.

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MIL-W-85/1	-	Waveguides, Rigid, Rectangular
MIL-W-85/2	-	Waveguides, Rigid, Rectangular (Heavy Wall)
MIL-W-85/3	-	Waveguides, Rigid, Rectangular (Millimeter Wavelength).
MIL-W-85/4	-	Waveguides, Rigid, Rectangular Reduced Height

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SPECIFICATIONS

MILITARY - Continued

MIL-P-116	-	Preservation-packaging, Methods of.
MIL-B-131	-	Barrier Materials, Watervaporproof, Flexible, Heat-Sealable.
MIL-S-13282	-	Silver and Silver Alloys.
MIL-B-22191	-	Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-C-45662	-	Calibration System Requirements.

STANDARDS

FEDERAL

FED-STD-356	-	Commercial Packaging of Supplies and Equipment.

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads 40" x 48" Pallets.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B107	-	Magnesium Alloy Extruded Bars, Rods, Shapes, and Tubes.
B210	-	Aluminum-Alloy Drawn Seamless Tubes.
B372	-	Seamless Copper and Copper-Alloy Rectangular Waveguide Tube

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B46.1	-	Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets.

3.2 Material. The material shall be as specified (see 3.1). Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Aluminum alloy. Aluminum alloys 1100, 6061, and 6063 shall conform to the chemical composition limits of ASTM B210. Alloy 1100 waveguides 3.000 by 1.500 inches and under in outside dimension shall be furnished in the H151 temper with a tensile strength of 17,000 pound-force per square inch (lb_f/in^2) minimum. Alloy 1100 waveguides over 3.000 x 1.500 outside dimension shall be furnished in the H112 temper with a tensile strength of 11,000 lb_f/in^2 minimum. Alloys 6061 and 6063 waveguides shall be furnished in the F temper.

3.2.2 Copper. Copper shall conform to type OF of DLP of ASTM B372.

3.2.2.1 Copper alloy. Copper alloy shall conform to commercial bronze, 90 percent, of ASTM B372.

3.2.3 Magnesium alloy. Magnesium alloy shall conform to the chemical composition and tensile property requirements for alloy AZ31B of ASTM B107.

3.2.4 Silver alloy. Silver alloy shall conform to the chemical composition requirements for grade C (coin silver) of MIL-S-13282. In addition, silver alloy waveguides shall be furnished in the drawn temper with a Rockwell hardness of 45.0 to 80.0 on the "B" scale.

3.2.5 Silver-lined copper or copper alloy. Silver-lined copper or copper alloy shall be copper (see 3.2.2) or copper alloy (see 3.2.2.1) laminated with a minimum of 0.005 inch of fine silver in accordance with grade A of MIL-S-13282.

3.2.6 Copper-clad invar. Copper-clad plating shall be of .003 to .007-inch thickness on both inside and outside surfaces.

3.3 Design and construction. Waveguides shall be of the design, construction, and physical dimensions specified (see 3.1).

3.3.1 Finish. The interior and exterior surfaces of the waveguide shall be bright, smooth, dry, and free of scales or oxide.

3.3.2 Length. The finished waveguide shall be furnished in 96, 120, 144, and 168 inches (8, 10, 12, and 14 feet) straight lengths (see 3.1) with ends. The shortest permissible length of the ends shall not be less than 60 percent of the nominal length (specific and stock), and the maximum permissible weight of ends shall not exceed 25 percent of the lot weight. Waveguide tube, ordered to specific or stock lengths, with or without ends (see 6.2), shall conform to the tolerances in table I.

3.3.3 Perpendicularity. Perpendicularity is the deviation from a right angle of two adjacent interior waveguide surfaces The deviation shall not exceed .008 inch per inch (1/2 degree) when measured at each end of the waveguide length.

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TABLE I. Length tolerances.
(Applicable only to full length pieces)

Length	Tolerance, inch (all plus)
Standard (stock)- - - - - - -	1
Specific	
Up to 14 ft, incl - - - - -	1/4
Over 14 ft - - - - - - - - -	1/2

3.4 Bow. When measured as specified in 4.6.2, the bow between any two points 2 feet apart on the concave external surface shall not be more than .010 inch on the narrow surface (E-plane) and .020 inch on the wide surface (H-plane) unless otherwise specified (see 3.1). If the waveguide is less than 2 feet long, the allowable bow shall be proportionate to the above requirements.

3.5 Twist. When waveguides are measured as specified in 4.6.3, the maximum twist along the longitudinal axis shall not exceed 1° per foot of length on the face of the interior or exterior surfaces.

3.6 Squareness of cut. When measured as specified in 4.6 4, the departure from squareness of the end of any waveguide shall not exceed 0.010 inch for waveguides up to 5/8-inch maximum inside waveguide dimension, inclusive, and 0.016 inch (1 degree) for waveguides over 5/8-inch maximum inside waveguide dimension.

3.7 Surface roughness. When measured as specified in 4.6.5, the interior surface roughness of the waveguide shall not exceed the values shown in table II.

TABLE II. Surface roughness tolerances of waveguides.

Specified major inner dimension (inches)	Allowable surface roughness, maximum arithmetic average (AA)	
	Aluminum, aluminum alloy and magnesium alloy	Copper, copper alloy, silver alloy, silver-lined copper, silver-lined copper alloy, and copper-clad invar
Up to 4, exclusive - - -	63	32
4 and over- - - - - - -	125	63

3.8 Pressurization (applicable to fabricated waveguides only, unless otherwise specified (see 3.1)). When assemblies are tested as specified in 4.6.6, there shall be no leakage from the edge seams of the waveguide.

3.9 Voltage standing wave ratio (VSWR) (applicable to fabricated waveguides only, unless otherwise specified (see 3.1)) When waveguides are tested as specified in 4.6.7, the VSWR shall not exceed 1.02.

3.10 Scratches. When measured as specified in 4 6.8, scratches shall not be more than 1 mil deep in the longitudinal direction of waveguides with internal measurements of 0 622 by 0 311 inch or larger. For waveguides less than 0 622 by 0.311 inch internal measurement, the depth of scratches shall not exceed 0 12 of the width.

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3.11 Marking. Each length of waveguide shall be marked in accordance with method I of MIL-STD-1285, with the part number (see 3.1) and the manufacturer's source code. The manufacturer's name or trademark may also be marked on the waveguide, provided such letters are not expressly forbidden in the contract or order, and their height does not exceed 50 percent of the outside dimension of the waveguide. The markings shall be applied with a permanent ink or paint. Not more than 6-inch intervals shall be permitted from the end of one marking to the beginning of the next marking.

3.12 Workmanship. The waveguide shall be free from defects of a nature that may interfere with normal military applications. The waveguide shall be uniform in composition and wall thickness, straight and smooth from end to end and shall be free from internal and external mechanical imperfections, and shall have a clean, bright appearance in accordance with good commercial practice. In addition, the interior surface of the waveguide shall be free from burrs, die marks, chatter marks, dirt, grease, scales, and splinters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.3).
b. Quality conformance inspection (see 4.5).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table III, used in fabricating the waveguides, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE III. Materials inspection

Material	Requirement paragraph	Applicable to specification or publication
Aluminum	3.2.1	ASTM B210
Aluminum alloy	3.2.1.1	ASTM B210
Copper	3.2.2	ASTM B372
Copper alloy	3.2.2.1	ASTM B372
Magnesium alloy	3.2.3	ASTM B107
Silver alloy	3.2.4	MIL-S-13282
Silver-lined copper	3.2 5	ASTM B372 and MIL-S-13282
Copper-clad invar	3.2 6	ASTM B372

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4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B, and C inspections.

4.5.1.1 Inspection lot. An inspection lot shall consist of all the waveguides of the same part number, produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of inspections specified in 4.6.1 and 4.6.1.1.

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality levels (AQL) shall be 1.0 and 4.0 (percent defective) for major or minor defects, respectively. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.2 2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table IV, in the order shown, and the sample shall be selected from inspection lots that have passed group A inspection.

TABLE IV. Group B inspection.

Examination or test	Requirement paragraph	Test method paragraph
Bow - - - - - - - - - -	3.4	4.6.2
Twist - - - - - - - - - -	3.5	4.6.3
Squareness of cut - - -	3.6	4.6.4
Pressurization 1/ - - -	3.8	4.6.6
VSWR 1/ - - - - - - - - -	3.9	4.6.7

1/ For fabricated waveguides, unless otherwise specified (see 3 1)

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 6.5 percent defective.

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract

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4.5.1.4 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table V, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

TABLE V. Group C inspection.

Examination or test	Requirement paragraph	Test method paragraph
Surface roughness- - - - - - -	3.7	4.6.5
Scratches- - - - - - - - - - -	3.10	4.6.8

4.5.1.4.1 Sampling plan. The first two lengths of the production lot shall be tested. Thereafter one out of each 25 lengths shall be tested.

4.5.1.4.2 Failures. If one or more sample units fail to pass group C inspection, the lot shall be considered to have failed.

4.5.1.4.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract.

4.5.1.4.4 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc, and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.2 Inspection of preparation for delivery Except when commercial packaging is specified, the sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination. Waveguides shall be examined to verify that the materials, design, construction, physical dimensions, finish, length, marking, perpendicularity and workmanship are in accordance with the applicable requirements (see 3.2 through 3.3.3, 3.11, and 3.12).

4.6.1.1 Dimensions. Waveguides with 1-inch major outside dimension and over shall be measured at a point at least 1 inch from the end; waveguides with less than 1-inch major outside dimension shall be measured at least 1/2 inch from the end to ascertain that all dimensions are as specified (see 3.1) and that the waveguide is of the specified length (see 3.1). Eccentricity may be measured at any point around the periphery. For the purpose of determining conformance with the dimensional requirements prescribed herein, any measured value outside the specified limiting values for any dimension shall be construed as failing to meet specification requirements

4.6.2 Bow (see 3.4) Bow is the departure from a straight longitudinal line connecting any two points 2 feet apart on the external surface of the waveguide without the effect of gravity or any other force The waveguide shall be so positioned during measurement that gravity will not tend to increase or decrease the amount of bow. Measurements shall be made along any 2-foot portion of the total length

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4.6.3 Twist (see 3.5). The angle of twist of waveguides shall be measured as shown on figure 1 and as follows:

a. The waveguide shall be placed on a flat, horizontal, reference plane. Waveguides with noticeable, flatwise, bow shall be positioned with the waveguide surface on the side of the convex curvature in contact with the reference plane.
b. One end of the waveguide surface shall be held in intimate contact with the reference plane by suitable means while the other end of the waveguide remains free.
c. Using an engraved, transparent, protractor segment, determine the angle line on the protractor which coincides to the nearest degree with the surface of the waveguide, while one edge of the protractor segment contacts, or is kept parallel to the reference plane by means of parallel face blocks.

4.6.4 Squareness of cut (see 3.6). The squareness of the cut of the waveguide shall be measured using an adjustable protractor.

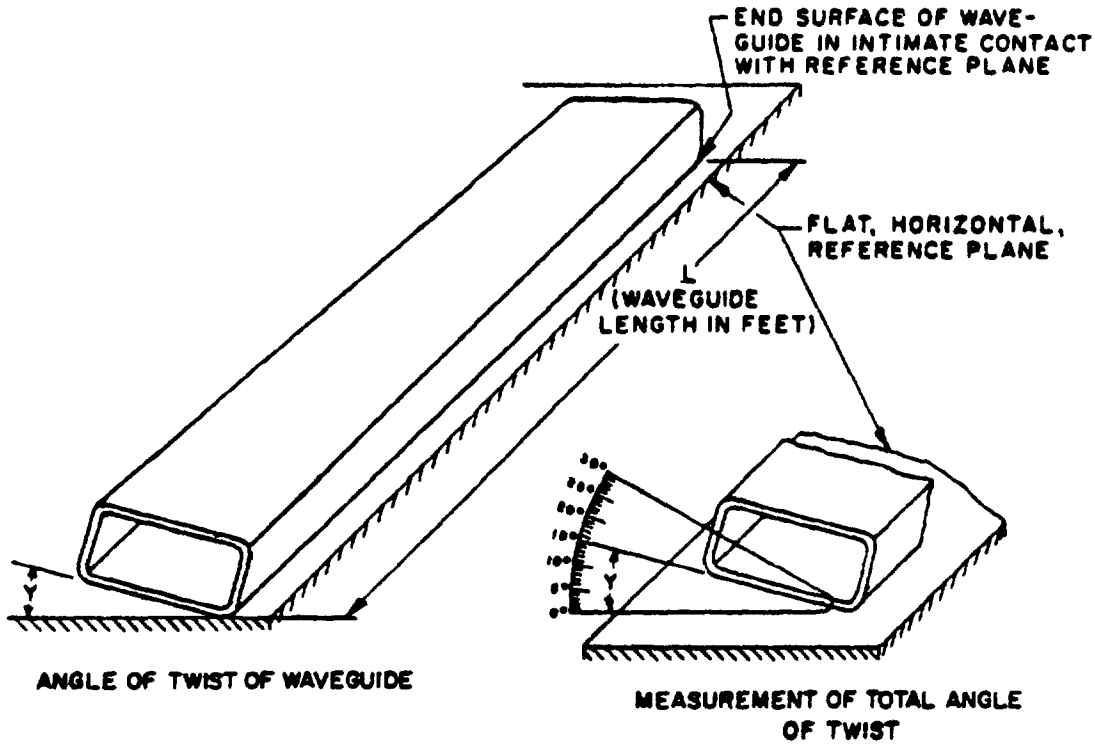
4.6.5 Surface roughness (see 3.7). The average interior surface roughness of the waveguide shall be measured in accordance with ANSI B46.1, except flaws shall be included in the surface roughness measurement and the corner radius shall not be included in the measurement. Surface roughness shall be measured across the grain over the full width of the waveguide interior surface, at the center of the waveguide length and 6 inches from the end of each length, unless otherwise specified (see 3.1 and 6.2).

4.6.6 Pressurization (see 3.8)(applicable to fabricated waveguides only, unless otherwise specified (see 3.1). The waveguide shall be subjected to the internal air pressure, in pounds-force per square inch (lbf/in2), specified in table VI while immersed in water for at least 5 minutes. The water temperature shall be 68° ±10°F. Any bubbles coming from the edge seams of the waveguide shall be considered as leakage. Bubbles that are the result of entrapped air on the exterior surface of the waveguide shall not be considered as leakage.

TABLE VI. Pressurization.

Inside width (nominal)	Wall thickness (nominal)	Pressure
inches		lb_f/in^2
23.000	0.188	0.25
21.000	0.188	0.25
18.000	0.125	0.25
15.000	0.125	0.25
11.500	0.125	0.50
9.750	0.125	0.50
7.700	0.125	1.0
6.500	0.125	1.5
5.100	0.080	2.0
4.300	0.080	3.0

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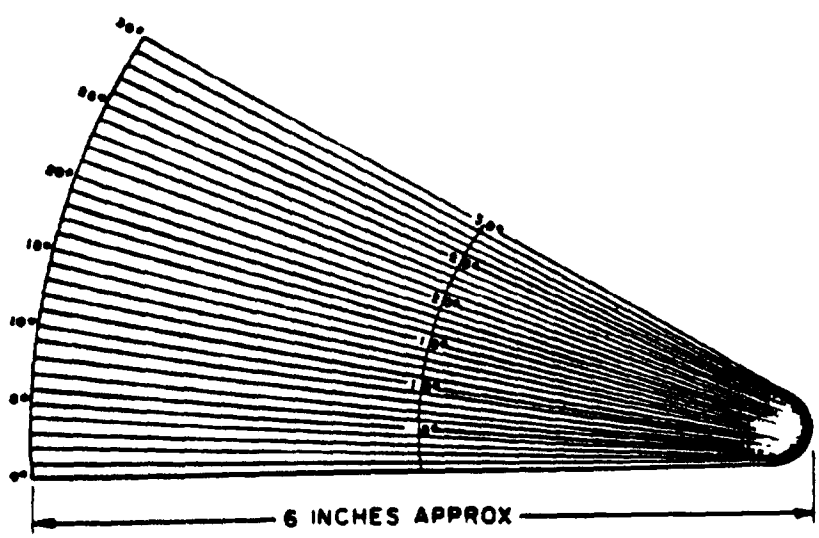


FIGURE 1. Method of determining twist.

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4.6.7 Voltage standing wave ratio (see 3.9)(applicable to fabricated waveguides only, unless otherwise specified (see 3.1)). The voltage standing wave ratio of the waveguide shall be measured over the frequency range specified in table VII The test equipment(s) shall be capable of providing a continuous measurement of VSWR over the required frequency ranges. A means shall be provided to produce a permanent record of the waveguides' VSWR versus frequency. If VSWR is not directly measured, that is, if return loss is measured and VSWR is calculated from that measurement, the permanent record shall indicate the worst case VSWR numerically for each frequency band and provide the calculation used to obtain the calculated VSWR. The measurement system and permanent record shall be capable of differentiating a VSWR change of at least 0.02.

TABLE VII. Frequency range.

Inside width (nominal)	Frequency range	Inside width (nominal)	Frequency range
inches	GHz	inches	GHz
23.000	0.32 - 0 49	9 750	0.75 - 1.12
21.000	0.35 - 0.53	7 700	0.96 - 1 45
18.000	0.410 - 0.625	6.500	1.12 - 1.70
15.000	0.490 - 0.750	5.100	1.45 - 2.20
11.500	0.64 - 0.96	4.300	1.70 - 2.60

4.6.8 Scratches (see 3.10). Examination for scratches shall be 6 inches from each end and at the middle of the length.

5 PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, or as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Waveguides shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Waveguides shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1 1.4 Unit packaging.

5.1.1.4.1 Waveguides 36 inches or less in length. Unless otherwise specified (see 6.2), waveguides 36 inches or less in length shall be individually packaged in accordance with MIL-P-116, submethod IA-8 insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Each unit package shall be placed in a supplementary container conforming to PPP-B-566, PPP-B-676, PPP-T-495, or PPP-B-636 except that wave guides exceeding 12 inches in length shall be restricted to supplementary containers conforming to PPP-B-636.

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5.1.1.4.2 Waveguides exceeding 36 inches in length. Unless otherwise specified (see 6.2), waveguides exceeding 36 inches in length shall be individually packaged in accordance with MIL-P-116, method IA as follows: The open ends of each waveguide shall be covered by a water-vaporproof barrier material conforming to MIL-B-131, class 1 or MIL-B-22191, type I. This barrier material shall extend a minimum of six inches back from each open end and shall be secured and sealed with tape conforming to PPP-T-66, type I. When the opening exceeds the width of the tape, cross strips shall be used to reinforce the barrier material. Alternately, the openings may be sealed with cover plates or caps and plugs secured with tape conforming to PPP-T-66, type I. All sealing devices shall meet the water-vaporproof requirements of MIL-B-131. The unit container shall conform to the level of packing specified (see 5.2).

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Waveguides shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. This level may conform to the contractor's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, or as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Waveguides 36 inches or less in length. The packaged waveguides shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1, or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix of PPP-B-636 using non-metallic or tape banding only.

5.2.1.2 Waveguides 36 to 72 inches in length. The packaged waveguides shall be individually packed in a fiberboard container conforming to PPP-B-636, class weather resistant, style optional, special requirements. Blocking and bracing shall be provided as necessary in accordance with MIL-STD-1186 to prevent movement of the waveguide within the container in lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement) requirements shall be applied in accordance with the appendix of PPP-B-636 using non-metallic or tape banding only.

5.2.1.3 Waveguides over 72 inches in length. The packaged waveguides shall be individually packed in a wood box conforming to PPP-B-601, overseas type, grade A or PPP-B-621, class 2, grade A. Blocking and bracing shall be provided as necessary in accordance with MIL-STD-1186 to prevent movement of the waveguide within the container. Closure and strapping requirements shall be as specified in the applicable specification.

5.2.2 Level B. Packing shall be as specified in 5.2.1.1, 5.2.1.2, and 5.2.1.3 except that the container shall conform to the domestic type or class of the respective specification.

5.2.3 Level C. The packaged waveguides shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity These packs shall conform to the applicable carrier rules and regulations and may be the contractor's commercial practice when such meets the requirements of this level.

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5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in contract, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Consolidation containers exceeding 200 pounds in gross weight shall have skids applied in a manner which will adequately support the items and facilitate the use of material handling equipment. Shipments of packed waveguides to a single destination shall be unitized by securely banding or strapping the packs together or by further packing in suitable consolidation containers. Unitized leads shall be uniform in size and quantities to the greatest extent practicable.

5.3 Marking. In addition to any special marking required by the contract (see 6.2), each unit package, supplemental and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.2.

5.4.3 Army procurements.

5.4.3.1 Level A (maximum military protection) unit packaging MIL-P-116, submethod IC-1 and method IC shall be used in lieu of submethod IA-8 and method IA, respectively. All supplementary containers shall either be overwrapped with waterproof barrier materials or shall conform to PPP-B-566 or PPP-B-676, variety 2 or 4 (see 5.1.1.4.1, 5.1.1.4.2 and 6.2).

5.4.3.2 Level A (maximum military protection) and level B (minimum military protection) packing. For level A packing the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2 style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I finish A. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1 and 5.2.2).

5.4.3.3 Level A and B unitization. For level A and B unitization, the fiberboard caps shall be weather resistant and softwood pallets conforming to NN-P-71, type VI, size 2 shall be used. The loads shall be bonded to the pallets by strapping conforming to QQ-S-781, type I, finish A or shrink film (see 5.2.4.1 and 5.2.4.2).

5.4.3.4 Commercial packaging. Commercial packaging (including unit packaging, packing and marking) shall be in accordance with FED-STD-356

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5.4.4 Navy procurements. The preparation for delivery requirements specified on Navy contracts apply only for direct Government shipments. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by Navy contractors or their authorized distributors shall be specified in the individual orders.

6. NOTES

6.1 Intended use. Waveguides covered by this specification are intended for use as transmission lines in military services radio and electronic equipment. For Naval applications, magnesium waveguides are for aeronautical use only.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. The complete part number (see 1.2 and 3.1).
c. Inspection of commercial packaging (see 4.5.2).
d. Levels of preservation-packaging and packing required (see 5.1 and 5.2).
e. Method of preservation, if other than submethod IA-8 or method IA (see 5.1.1.4.1, 5.1.1.4.2, and 5.4.3.1).
f. Special marking, if required (see 5.3).
g. Stock lengths (with or without ends) or specific lengths of waveguides required (see 3.3.2).
h. Special methods or increased frequency of checking surface roughness (see 4.6.5).

6.3 Engineering information. Additional engineering information for waveguides is available in MIL-HDBK-216, "RF Transmission Lines and Fittings," copies are available upon request from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement NATO STANAG 4035. When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required. The United States, by international agreement (STANAG), has agreed to the use of the types of waveguides designated by NATO nomenclature, ie, NWG-XXXX, in new equipment design. These NATO types are shown in the associated specification sheet, and should be used whenever possible.

Custodians:
Army - EL
Navy - EC
Air Force - 85

International agreements (see 6.4)

Review activities:
Army - MU, MI, SG, ME
Navy - OS, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - AT, MC
Navy - MC, AS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0819)

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3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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